TA14 | Healthcare Evidence to Action

Blue IoT / Encompass Blue discussion surface — independent illustrative technical examination

Purpose. Demonstrate how a hospital HVAC proposal can be examined for admissible evidence, applicable authority, established standing, risk-proportionate approval, authorized execution, and verified outcome. This is a conceptual TA14 demonstration, not a deployment, endorsement, integration, clinical recommendation, or operating procedure.

The governing question

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

Illustrative case

A CO■ sensor reports sustained elevation in an occupied general patient ward. An engineer proposes a bounded outdoor-air adjustment. The ward, reading, equipment, limits, and approvals depicted in the companion visual are fictional training values, not measurements from Blue IoT or a hospital. CO■ alone does not establish ventilation adequacy, infection risk, cause, or a safe control action.

The eight examination stages

01 Reality / signal — Identify the observation, affected zone, patient-care context, uncertainty and competing explanations.

02 Record / continuity — Preserve raw values, device identity, calibration and quality history, time synchronization, location attribution, and derivations separately.

03 Proposition — Specify the exact proposed change, equipment, limits, duration, clinical dependencies, rollback criteria and expected effect.

04 Authority / standing — Determine the governing site policies and the actual actor, capacity, standing basis and authority relationship for this action NOW.

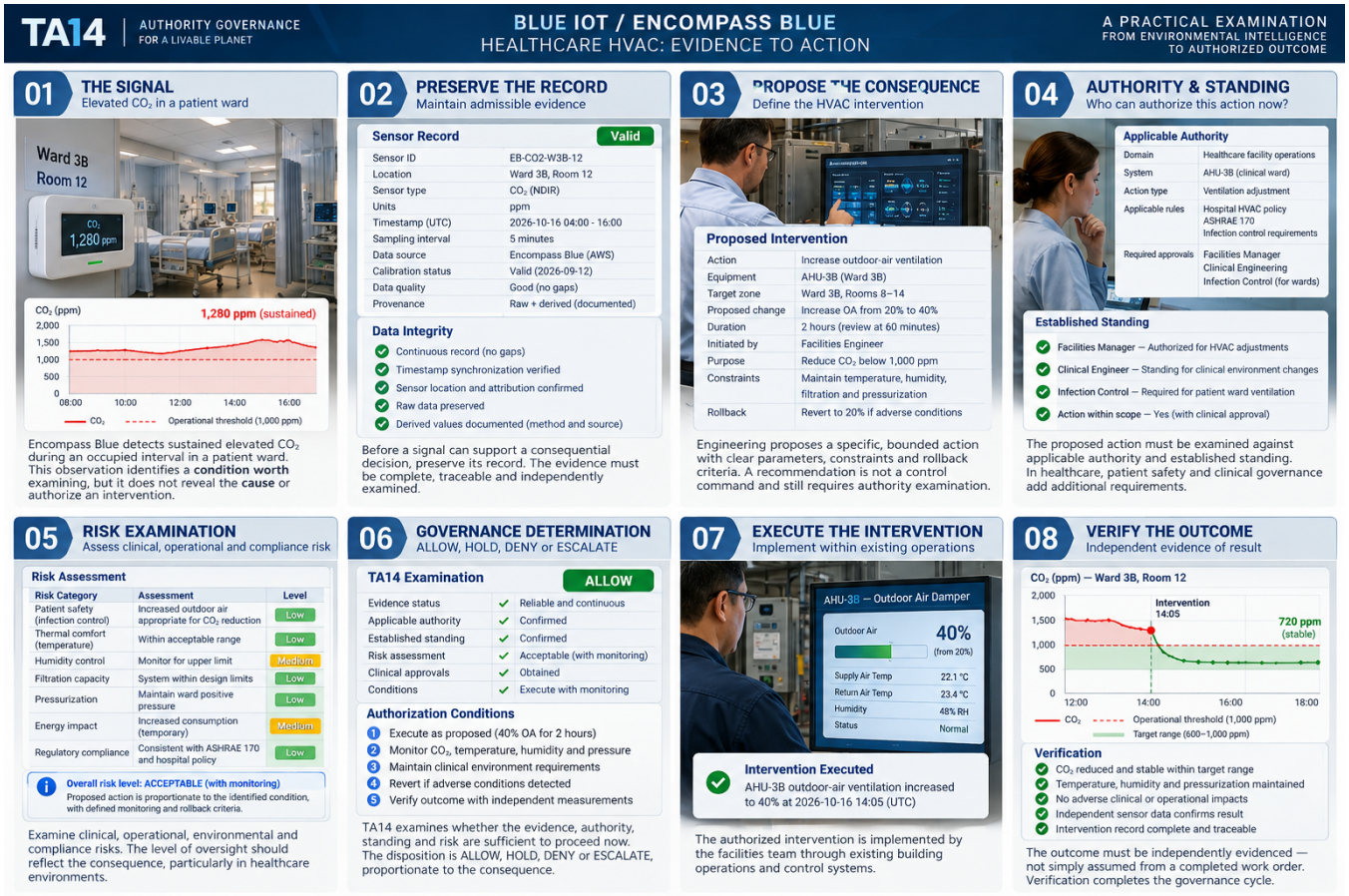
05 Risk / admissibility — Examine clinical environment, infection prevention, pressure relationships, humidity, filtration, system capacity, energy, and patient impact.

06 Binding determination — Return ALLOW, HOLD, DENY or ESCALATE with reason, scope, validity, required approvals and evidence references.

07 Commit / execution — Authorized local operators or BMS apply independent safety constraints; log actual command, acknowledgment, and observed equipment state.

08 Outcome — Compare post-action measurements and clinical/operational constraints; preserve uncertainty and distinguish execution from verified benefit.

Companion visual — eight-stage overview



All values, timestamps, roles, approvals, and outcomes in this visual are illustrative. The ALLOW branch is conditional; no real authority, patient safety validation, or physical execution has been established.

Risk-proportionate decision model

LOWER CONSEQUENCE: Noncritical, reversible change inside approved parameters with validated evidence and delegated standing may qualify for a bounded ALLOW. Site rules remain controlling.

HIGHER CONSEQUENCE: Patient-care zones, airborne isolation, operating theatres, critical pressure relationships, life-safety functions, or uncertain clinical impact require specialist review and often ESCALATE. A single CO■ reading is insufficient for a ventilation change.

HOLD: Evidence gap, missing standing, missing site approval, stale context, or unresolved equipment/clinical constraint that could be remediated.

DENY: Proposed action conflicts with applicable prohibition or established safety requirement, or exceeds non-delegable constraints.

ESCALATE: Risk or ambiguity exceeds the bounded decision authority; route to authorized clinical, infection-prevention, facilities and engineering personnel as site policy requires.

Illustrative decision branches

Branch	Condition	Disposition
A	Evidence traceable; approvals and standing confirmed; all constraints validated	Conditional ALLOW
B	Reliable readings; authority relationship not established	HOLD
C	Proposal would violate required pressure relationship	DENY
D	Clinical implications or infection-control effects unresolved	ESCALATE

Minimum evidence and governance interface

Input record — Sensor identifier; raw and derived measurements; timestamps; provenance; calibration/quality flags; room and equipment mapping; contextual occupancy and operating conditions.

Proposed consequence — Actor; action type; target system and zone; duration; allowable bounds; expected effect; hazard analysis; rollback and expiry.

Authority evidence — Applicable site policy; required human approvals; identity, role, delegated capacity, standing basis, scope and effective period.

TA14 response — Disposition; reason codes; admissibility findings; policy/standing references; bounded scope; conditions; expiry/revalidation triggers; immutable decision record.

Execution feedback — Local BMS/operator acceptance or rejection; command trace; actual equipment state; exceptions and override reasons.

Outcome evidence — Post-action trends, sensor quality, equipment states, environmental and clinical constraint checks, independent verification and unresolved uncertainty.

What to test with Blue IoT

1. Which Encompass Blue observations and device metadata are available without assuming an API or integration?

2. Which existing work-order, BMS, facilities and clinical approval systems establish local execution authority?

3. Which high-consequence cases require a human checkpoint rather than machine-speed automatic disposition?

4. Can a bounded governance response be recorded without TA14 issuing any building control command?

5. Can post-action evidence demonstrate actual effect while preserving confounders and uncertainty?

Boundary statement

TA14 is an independent proposed governance layer. Blue IoT and Encompass Blue are discussed as potential environmental intelligence sources only. No partnership, adoption, certification, endorsement, integration or transfer of execution authority is implied.